

Kunj Rathod

kunjrathod.com +1 (385) 202-8879 kunj.rathod@utah.edu linkedin.com/in/rathodkunj github.com/rathodkunj2005

Education

University of Utah

Aug 2023 – Dec 2026

B.S. in Computer Science

GPA: 3.7/4.0, Dean's List

Coursework: Distributed Systems, Operating Systems, Machine Learning, Computer Vision, NLP, Algorithms, Databases

Research

MEMROT v2 — Mechanistic Anatomy of Long-Context Memory Failure

May 2026 – Present

Independent research artifact; Gemma-2-2B, Gemma Scope SAEs, LongMemEval-S, CHPC/SLURM

- Built a reproducible mechanistic-evaluation pipeline for controlled haystack retrieval at fill levels $k \in \{0, 1, 2, 4, 8, 14\}$: gold-position randomization, in-hook attention reduction, empirical SAE layer scans, mean-ablation/intervention tests, and an 840-pass sweep over 140 exactly-one-gold questions.
- Framed the paper as a single-model case study with explicit limitations around feature selection, attention normalization, and heuristic grading, prioritizing honest failure analysis over inflated claims.

Disentangling Correct Refusal from Over-Refusal via Sparse Feature Interventions

Mar 2026 – May 2026

Qwen2.5-0.5B-Instruct; sparse autoencoders; refusal-safety evaluation

- Built a 400-example unsafe/benign-overlap benchmark and trained sparse autoencoders on residual activations to rank refusal-feature families; causal steering of one family improved held-out unsafe refusal from 0.80 to 0.85, while suppressing top over-refusal features did not reliably improve benign compliance — a clean negative result.

Circuit-Level Generalization in LLMs via Sparse Feature Circuits

Jan 2026 – Present

University of Utah; attribution graphs, sparse feature circuits, graph skeletonization

- Building automated pipelines to extract sparse feature circuits and predict transfer to non-default tasks via attribution graphs and Pathways Discovery; open-sourced a skeletonization tool that prunes graphs into supernodes through Mapper embeddings while preserving source metadata.

Long-term Active Embodied Question Answering in Simulated Indoor Environments

Spring 2026

University of Utah; embodied AI, VLM video QA, episodic memory, SceneSmith, MuJoCo

- Built Video Mind Palace, a sampled-frame VLM baseline for LA-EQA that replaces scene-graph memory; cut online inference time by 31–57% versus Robotic Mind Palace with a 13–17% relative accuracy tradeoff, then proposed SceneSmith + MuJoCo environments for longer interactive embodied QA.

Industry Experience

Software Engineering Intern — Microsoft Fabric (Azure Data)

May 2026 – Aug 2026

Microsoft, Redmond, WA

- Building SKU Change History end-to-end (query builder, service layer, REST API) for the Fabric Manage Capacities admin portal with telemetry-backed audit timelines; using Microsoft's internal AI stack to accelerate debugging and implementation planning.

Software Development Intern, AI Services (SUDO Program)

Jan 2025 – Present

University of Utah Health, Salt Lake City, UT

- Architected HIPAA-compliant LLM serving infrastructure for 90+ hospital executives: token-streaming inference at p95 <200ms TTFT and 40% latency reduction; owned AWS Bedrock Agents, Knowledge Bases, and Guardrails plus session persistence for 1,000+ concurrent conversations.

AI Engineering Intern

Nov 2024 – Apr 2025

CourtEasy.ai / Nugen (Remote)

- Scaled hybrid retrieval (dense vectors + BM25 + reranking) over 10M+ documents (+28% recall, -35% hallucinations); benchmarked 8 LLM families on F1, latency, and cost and built model routing that cut inference spend.

Selected Projects

Continuum — Local-First Multimodal Memory Engine

Jun 2026

Rust, Tauri 2, React, TypeScript, LanceDB, OCR, embeddings, MCP

- Built a privacy-first desktop memory layer capturing foreground context on-device into vector + lexical retrieval fields in LanceDB, exposing agent-ready recall through an MCP server.

HirePilot — Autonomous AI Recruiting Pipeline (Podium AI Hackathon Winner)

Mar 2026

Python/FastAPI, Anthropic, Perplexity, GitHub REST, Twilio, Slack, Google Calendar

- Shipped a real-integration 9-agent recruiting loop in 24 hours: GitHub sourcing, Perplexity enrichment, Anthropic ranking and outreach, Twilio live voice interviews, and manager approval controls.

Languages & Technologies

AI/ML: PyTorch, TensorFlow, JAX, vision-language models, video QA, embodied AI, agent evaluation, RAG/GraphRAG, LangChain, LlamaIndex

Systems & Cloud: AWS (Lambda, API Gateway, S3, DynamoDB, RDS, Bedrock), Azure, Docker, Kubernetes, FastAPI, gRPC, CI/CD, SLURM, distributed systems

Languages: Python, C++, Java, Rust, Swift, TypeScript, SQL